

A Profitable Strategy for Short-Term Traders

How to use VIX Index for Mean-Reverting SPY Opportunities



CONCRETUM RESEARCH

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Markets rarely fall in calm, orderly ways; as the old Wall Street adage goes, they *“take the stairs up and the elevator down.”*

We also know that during downside moves, **technical and behavioral forces can trigger positive feedback loops** that exacerbate market sell-offs.

Investors cut risk, volatility-managed strategies deleverage, hedgers chase protection, and the price of **downside insurance** rises faster than fundamentals alone would seem to justify. In other words, **downside moves are often made worse by the way investors react to them.**

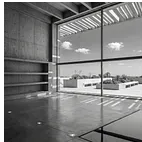
At the same time, speculators who have kept capital available for richer opportunities can step in, provide liquidity to forced sellers, and try to exploit abnormal corrections.

As quantitative traders, we often look to **implied volatility to identify short-term panic regimes**. The VIX, commonly referred to as the **“fear gauge”**, is the best-known measure of this kind: in technical terms, it reflects the level of implied volatility embedded in S&P 500 option prices, providing a market-based estimate of expected volatility over the next month.

In several of our previous studies, we have shown how **investors tend to overpay for such protection**. This creates profitable trading

opportunities in VIX-linked instruments and derivatives...

One example is documented in our paper “The Volatility Edge”, which recently placed fifth in the 2026 Quantpedia Awards.



The Volatility Edge

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... but we think there is another way to exploit the VIX.

For instance, we can use it to identify when equity markets enter a **temporary stress regime**, when moves tend to become over-amplified by the technical and behavioral forces described above, and then implement the trade directly **through SPY**.

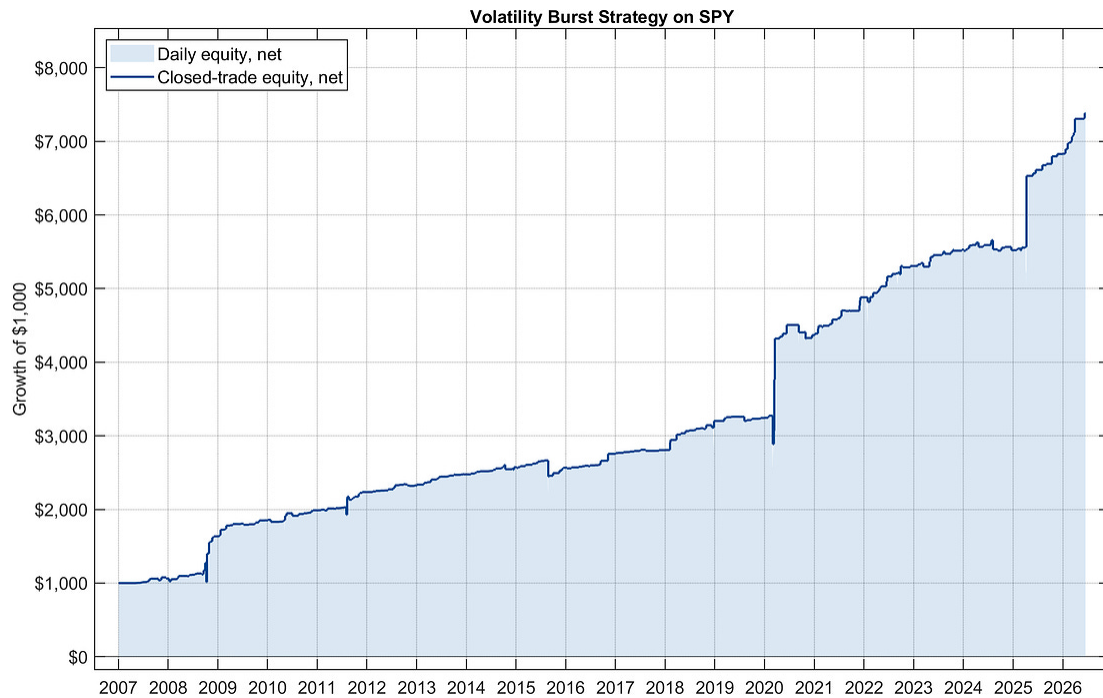
In this article, we present a simple system built around **three core components**:

1. A VIX-based signal to **identify volatility bursts**;
2. Position-sizing rules to **exploit prolonged weakness**;
3. Exit criteria designed to **capture a potential market rebound**.

From January 2007 through June 2026, our framework generated a **10.8% CAGR net of fees**, while being active on only **14%** of trading days.

The strategy does not require **continuous intraday monitoring** or **discretionary intervention**: signals are computed shortly before the close and implemented via **market-on-close (MOC) orders**, resulting in a relatively simple operational workflow.

In this article, we walk through how the signal is defined, how exposure is increased when stress persists, how the same-day implementation avoids look-ahead bias, and where the approach's strengths and weaknesses appear in the data.



We also view this as the **first step in a broader research project**. The results are promising enough that, over the next few weeks, we plan to publish a follow-up piece applying the same methodology across multiple assets and implied-volatility indexes, along with **Python code** to help readers automate the strategy via **Interactive Brokers APIs**.

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Note for the reader. *This article presents the findings of an empirical research study based on historical market data. The results are derived from backtests and therefore reflect past market behavior, which may differ materially from future market conditions. While historical analysis can provide useful insights into systematic trading strategies, it should not be interpreted as a guarantee of future performance or as investment advice. Readers are encouraged to evaluate the methodology critically and draw their own conclusions regarding its potential applicability.*

The VIX as a Signal

The core idea is rather simple.

When implied volatility **moves unusually far above its short-term range** and SPY is simultaneously under pressure, the strategy enters long SPY near the close with a pilot position. Then (if stress persists) the system can add exposure, and once SPY rebounds, the position is liquidated.

This makes the setup distinct from a static equity allocation: it is a **capital-light, episodic exposure** to moments when abnormal fear may have pushed equity prices too far.

In our context, the key point is that the VIX is a **market price for expected volatility**, and that price is strongly affected by how investors price downside risk.

Importantly, we are **not buying or shorting VIX futures, VIX ETPs, or variance products**: we are using the VIX only as a signal to identify equity-market conditions in which **fear may have moved too far, too quickly**.

The trade itself is executed through SPY, the most liquid ETF tracking the S&P 500, even though the same idea could also be implemented via ES or MES futures.

Defining a Volatility Burst

Each day, 15-minutes before the close, we define an **upper band** for the VIX Index through the following steps.

First, we compute a 5-day simple moving average:

$$\text{SMA}_5(\text{VIX})_t$$

Next, we quantify how much the VIX has fluctuated around this short-term average over the past 21 days by computing the standard deviation of the spread between the VIX and its 5-day SMA:

$$\text{BandWidth}_t = \text{Std}_{21}(\text{VIX}_t - \text{SMA}_5(\text{VIX})_t)$$

We then compute the upper volatility band as:

$$\text{UpperBand}_t = \text{SMA}_5(\text{VIX})_t + 1.0 \times \text{BandWidth}_t$$

We identify a volatility-burst day when today's VIX is above this upper band. The magnitude of the burst is:

$$\text{VolatilityBurst}_t = \text{VIX}_t - \text{UpperBand}_t$$

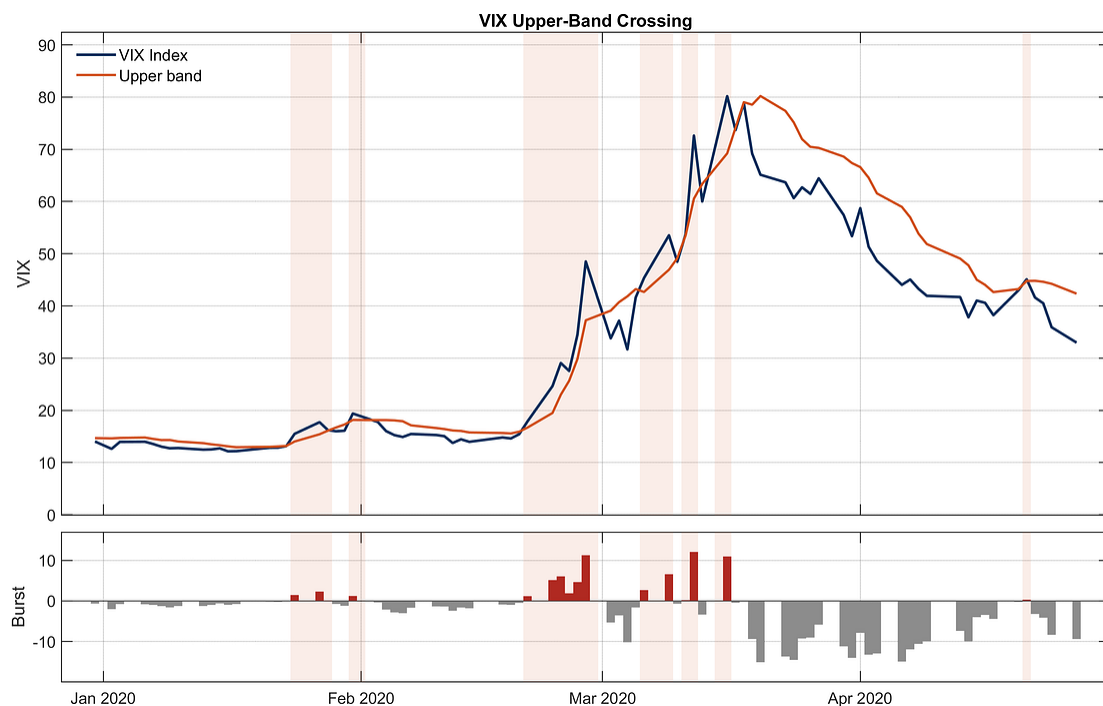
Finally, a burst is detected when:

$$\text{VolatilityBurst}_t > 0$$

In plain words, the signal activates once the VIX moves above this upper band level.

It is worth noting that this indicator is **not designed to forecast volatility**. Rather, it is meant to **identify a market state** in which implied volatility has risen unusually high relative to its recent behavior.

A visual example is provided in the chart below, using the COVID selloff in early 2020. The blue line shows the VIX Index, while the orange line marks the upper volatility band: when the VIX moves above that band, the volatility-burst signal turns positive. The lower panel reports the value of the burst itself, with positive readings highlighted in red.



Entry Conditions

A long SPY position is initiated only when the volatility burst coincides with equity weakness.

The entry conditions are:

1. Volatility **burst signal is active**;

2. **SPY is down** from the prior close;
3. SPY is trading in the bottom 20% of its recent 5-day range.

The second and third conditions ensure that the VIX event is confirmed by the price action of SPY itself. **We do not want to buy on every unusual VIX move:** we want to buy when the options market is signaling abnormal stress and the underlying equity market is confirming that stress through weakness of its own.

Operationally, the signal is computed **15 minutes before the close**. On normal trading days, this means 15:45 New York time; on early-close days, the equivalent signal time is 12:45. The assumed execution is a **market-on-close order**, or MOC order, designed to execute at the **official closing auction price**: on normal trading days, that auction occurs at 16:00 New York time; on early-close days, it occurs at 13:00.

This is important for two reasons.

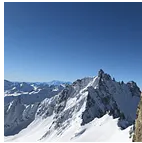
First, it makes the backtest **more realistic**. A less careful test might wait for the official daily close, observe all needed information, compute the signal using that information, and then assume instant execution. This procedure would not be directly implementable: we avoid this issue by **computing the signal before the closing auction**, using only information that would have been available at that time.

A practical note. If you use daily OHLC data for the VIX to generate trading signals that are applied to SPY (or other US ETFs), be aware that the reported VIX closing price is based on option prices observed at **4:15 p.m. ET**, whereas the U.S. equity market closes at **4:00 p.m. ET**. As a result, generating a signal from the daily VIX close and assuming execution via a same-day market-on-close order introduces (at least) a **15-minute look-ahead bias**. Our implementation avoids this issue by

computing the signal at **3:45 p.m. ET**, using only information that was actually available before the equity market closed.

Second, it **preserves the economic timing of the trade**. The strategy is designed to enter during the same session in which stress is observed, not the next morning: by computing the signal before the close, the model leaves enough time to submit market-on-close orders.

This links to a broader point we have discussed in a recent study: for a deeper look at how execution timing can play a fundamental role in implementing short-term signals, we recommend reading the article below.



Alpha Expires While You Sleep

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Adding Into a Mean-Reverting Trade

We attempt to keep the position-sizing approach deliberately simple and aligned with the mean-reverting nature of the trade.

Each initial entry is small. Then, if a trade remains active and the market continues to weaken, the system is allowed to add exposure, subject to a **maximum leverage cap of 3x** (in our tests, that limit was reached only on a handful of occasions).

Formally, for the first entry:

$$w_t = \min(\text{MaxLeverage}, 0.25 \times \max(1, \lfloor \text{VolatilityBurst}_t + 1 \rfloor))$$

And for additional entries while the trade remains open:

$$w_t = \min(\text{MaxLeverage}, w_{t-1} + 0.25 \times \max(1, \lfloor \text{VolatilityBurst}_t + 1 \rfloor))$$

The position increases in discrete 0.25x steps, scaled by a coarse measure of burst intensity and subject to a maximum exposure cap. Importantly, the system can add only if the position remains open and the exit condition has not been triggered; in other words, **it does not add blindly or indefinitely**, but only while the trade remains in the same stress-reversal state.

This kind of “**add into losers**” logic is often dismissed as dangerous. Used blindly, it can be, but within a mean-reversion framework, this philosophy is **not inherently flawed**. The important questions are:

*Does the strategy exploit a **reliable statistical edge**?*

*Is the pilot position **small enough** to allow multiple adds?*

*Is the exit condition **easy enough** to reach?*

*Is leverage **capped**?*

Here, the logic is that if the stress episode continues, the system can improve its average entry price while the position remains in the same mean-reversion regime.

While in this article we **deliberately stick to a simple sizing approach**, other methods could make exposure depend on realized volatility, drawdown, liquidity, option-implied skew, or other nonlinear measures of stress.

Note - The sizing formula is not the source of the edge (similar results can be obtained with several alternative sizing schemes). We use this implementation simply as a transparent baseline.

Exit Rule

In our implementation, a position is liquidated when either of the following conditions is met:

1. **SPY** posts a **positive return from the previous close**,
2. **SPY** is **no longer** in the **lower 20% of its recent 5-day range**.

The second condition uses the same range measure used in the entry filter: if SPY has moved out of the bottom quintile of that short-term range, the stress state is considered to have eased enough to close the position.

Notably, **this matters for leverage as well**. A 3x cap is substantial, but risk depends not only on the leverage multiple itself; it also depends on how quickly the strategy can exit once the initial shock begins to fade. A relatively easy-to-trigger exit rule helps **reduce the risk of remaining exposed too long, and with too much leverage**, after the original stress event has passed.

Historical Performance

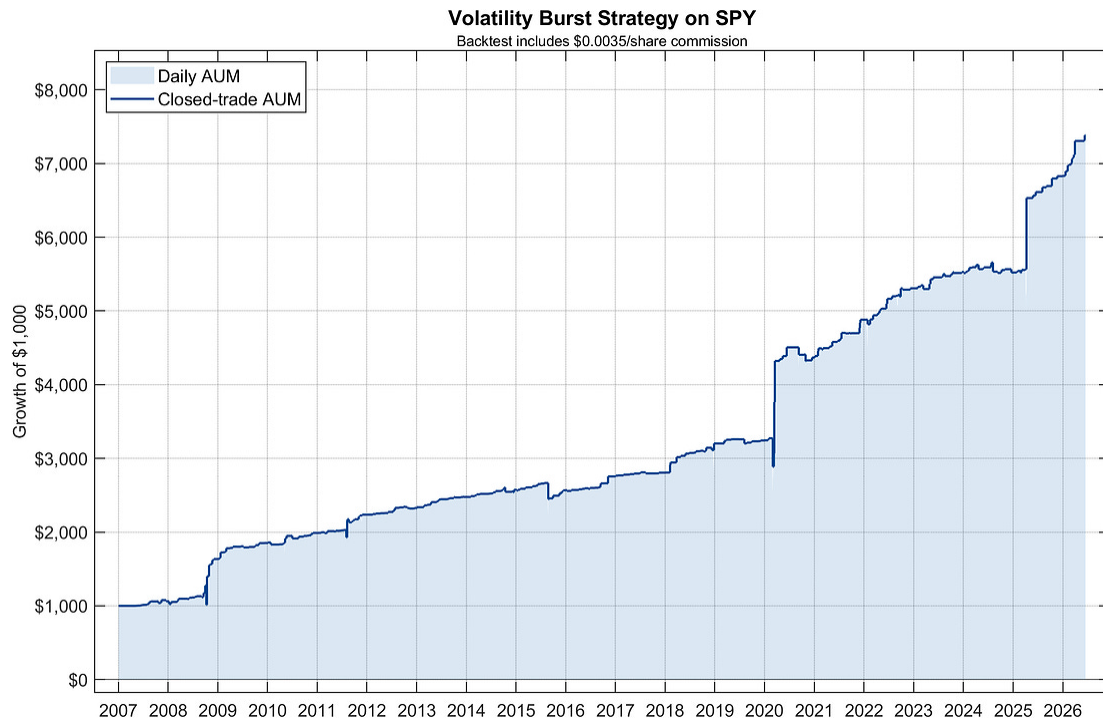
The tested sample runs from January 2007 through June 2026, and the results below are shown **net of transaction costs** (assuming \$0.0035 per share applied on entries, adds, reductions, and exits), and they exclude financing costs, slippage, and interest earned on idle cash.

The following tables and charts evaluate the strategy from three perspectives: portfolio-level performance, month-by-month returns, and trade-level behavior.

Equity-Level Results			
Total Return	638.2%	CAGR	10.8%
Volatility	20.0%	Sharpe Ratio	0.60
Sortino Ratio	1.55	Adjusted Sharpe	1.28
Max Drawdown	23.5%	Days Invested	14.0%

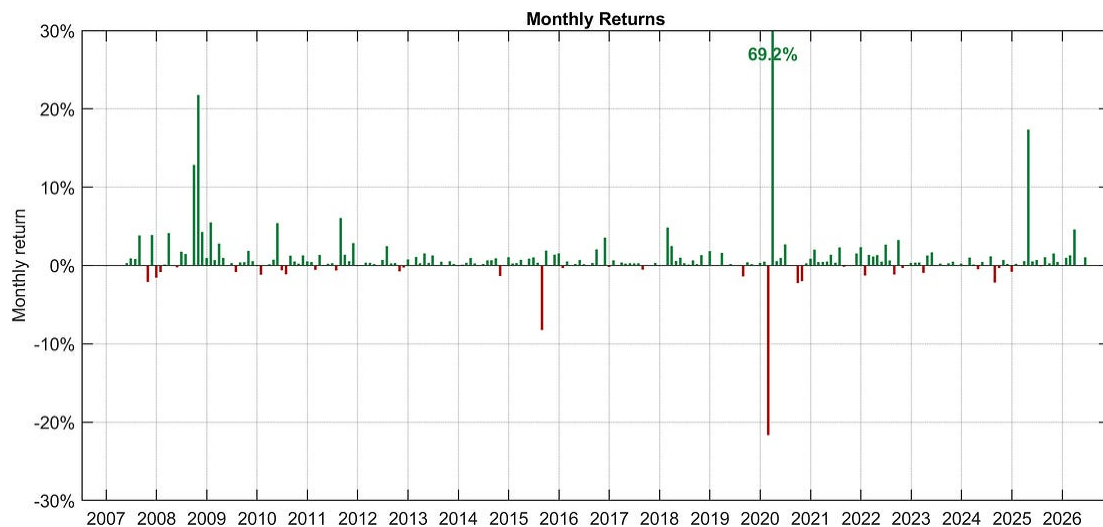
The results are notable: the strategy generated a **10.8% CAGR net of costs**, essentially matching the 10.9% CAGR of a SPY buy-and-hold position over the same period, while being invested on only 14% of trading days (*still, this framework is not meant to be a direct substitute for an “always-on” equity exposure*).

Because our signal is active only episodically, we also report an **adjusted Sharpe ratio measure computed only on days when the net strategy return is nonzero** (then annualized by multiplying by $\sqrt{252}$). This is not a replacement for the canonical Sharpe ratio, but it helps us isolate the return profile of days when capital is actually deployed.



Monthly Performance

The tables below report both monthly and yearly portfolio returns **net of transaction costs**. Out of 234 calendar months, 165 were positive and 36 were negative, with the remaining months essentially flat, while at the calendar-year level, 18 out of 20 years showed positive returns.



The best year was 2008, with a return of 54.1%, while the worst year was 2015, at -0.2%. The best month was March 2020, at 69.2%, and the worst month was February 2020 (-21.7%); remarkably, **both observations occurred during the COVID sell-off.**

Monthly Performance													
Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
2007	0.0%	0.0%	0.0%	0.0%	0.3%	0.9%	0.8%	3.8%	0.0%	-2.1%	3.9%	-1.5%	6.1%
2008	-0.8%	0.1%	4.1%	0.0%	-0.2%	1.8%	1.5%	0.0%	12.8%	21.8%	4.3%	1.0%	54.1%
2009	5.5%	0.7%	2.8%	1.0%	-0.1%	0.3%	-0.8%	0.4%	0.4%	1.9%	0.5%	0.1%	13.3%
2010	-1.2%	0.1%	0.2%	0.8%	5.4%	-0.6%	-1.1%	1.3%	0.5%	0.2%	1.3%	0.5%	7.4%
2011	0.5%	-0.6%	1.4%	0.0%	0.2%	0.3%	-0.6%	6.1%	1.4%	0.5%	2.9%	0.0%	12.5%
2012	0.0%	0.4%	0.3%	0.2%	0.0%	0.7%	2.5%	0.3%	0.3%	-0.8%	-0.3%	0.8%	4.5%
2013	0.0%	1.1%	0.3%	1.5%	0.3%	1.3%	0.0%	0.5%	0.0%	0.5%	0.2%	0.1%	5.9%
2014	0.1%	0.3%	1.0%	0.3%	0.0%	0.2%	0.7%	0.7%	0.9%	-1.3%	0.0%	1.1%	3.9%
2015	0.3%	0.3%	0.7%	0.0%	0.9%	1.0%	0.4%	-8.2%	1.9%	0.0%	1.4%	1.5%	-0.2%
2016	-0.3%	0.5%	0.0%	0.2%	0.7%	0.2%	0.0%	0.3%	2.1%	0.0%	3.6%	-0.2%	7.1%
2017	0.7%	0.0%	0.4%	0.2%	0.3%	0.3%	0.3%	-0.5%	0.0%	0.0%	0.3%	0.1%	2.1%
2018	0.0%	4.9%	2.5%	0.6%	1.0%	0.3%	0.1%	0.6%	0.2%	1.3%	0.0%	1.8%	14.1%
2019	0.0%	0.0%	1.6%	0.0%	0.2%	0.0%	0.0%	-1.4%	0.4%	0.2%	0.1%	0.3%	1.3%
2020	0.5%	-21.7%	69.2%	0.6%	1.0%	2.7%	0.0%	0.0%	-2.2%	-2.0%	0.3%	0.9%	34.6%
2021	2.0%	0.4%	0.5%	0.5%	1.4%	0.4%	2.3%	-0.2%	0.0%	0.0%	1.5%	2.3%	11.8%
2022	-1.3%	1.4%	1.1%	1.3%	0.5%	2.7%	0.6%	-1.1%	3.3%	-0.3%	0.0%	0.3%	8.7%
2023	0.4%	0.4%	-0.9%	1.3%	1.7%	0.0%	0.2%	0.1%	0.3%	0.5%	0.0%	0.2%	4.2%
2024	0.0%	1.0%	0.1%	-0.5%	0.5%	0.0%	1.2%	-2.2%	-0.3%	0.7%	0.2%	-0.8%	-0.1%
2025	0.2%	0.0%	0.6%	17.4%	0.5%	0.7%	-0.1%	1.1%	0.3%	1.5%	0.5%	0.0%	23.7%
2026	1.0%	1.3%	4.6%	0.0%	0.0%	1.0%							8.1%

Trade-Level Results

An average (gross) trade return of **55 bps**, together with an average winning trade of **94 bps** and an average losing trade of **-113 bps**, suggests that there is **sufficient room to absorb realistic implementation costs**, especially given that the strategy can be implemented through highly liquid instruments (such as SPY or index futures) and trades infrequently.

Trade-Level Results			
Number of Trades	398	Trades per Year	20.5
Hit Ratio	80.9%	Average Return	55 bps
Average Win	94 bps	Average Loss	-113 bps
Average Duration	1.7 days	Average Adds	0.7
Average PnL/share	\$0.66	Median PnL/share	\$0.77
Best Trade	3,834 bps	Best Trade Exit	2008-10-13
Worst Trade	-2,059 bps	Worst Trade Exit	2008-10-08

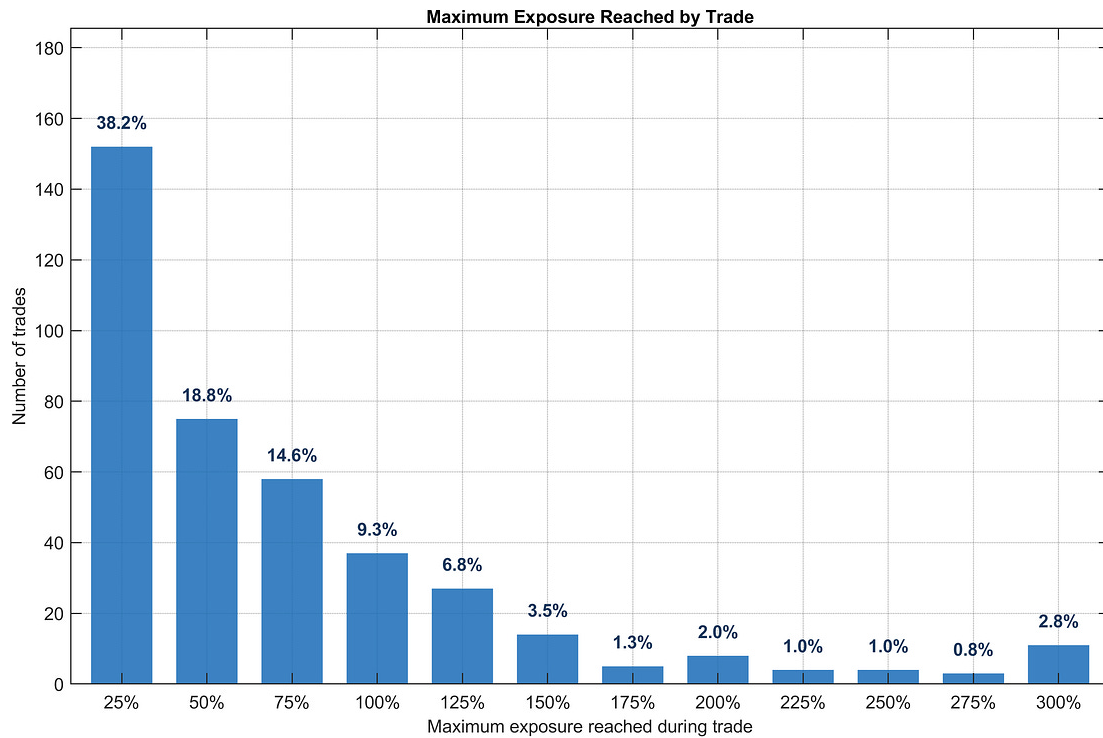
Trade-level figures are presented gross-of-costs

The Role of Leverage

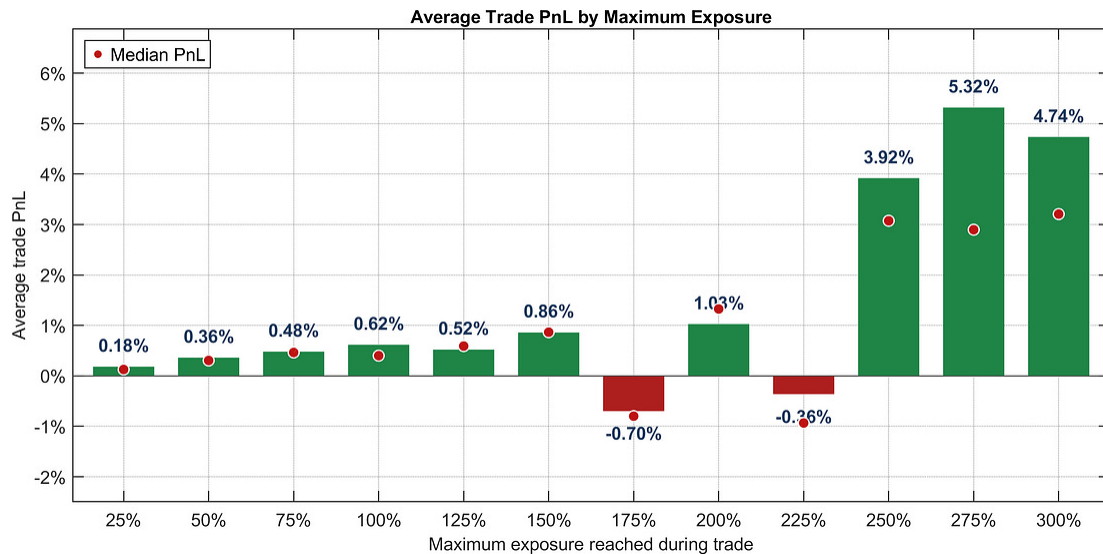
One aspect that deserves clarification is the role of leverage.

At first glance, a maximum exposure of **3x** may appear aggressive. In practice, however, such levels are reached only rarely because exposure is increased only while a trade remains active and fresh volatility-burst signals continue to be generated.

The chart below reports the maximum exposure reached during every completed trade. Most positions never require leverage at all: **80.9% of all trades remained at or below 100% exposure**, while only **19.1%** ever exceeded 1x. Trades reaching **2x or more represented just 7.9% of the sample**, and the full **3x** exposure was reached in only **2.8%** of all trades.



The next figure reports the average trade return conditional on the maximum exposure reached during the trade. Interestingly, the highest average returns are observed among the relatively small subset of trades that required larger exposure. This is consistent with the economic intuition behind the strategy: **the strongest volatility bursts often coincide with periods of exceptional market stress, which subsequently offer the greatest mean-reversion opportunities.**



At the same time, these results suggest that the strategy is **not critically dependent on the 3x leverage cap**. Since only a small fraction of trades ever reaches the highest exposure levels, investors adopting a lower leverage limit would likely still capture a substantial portion of the strategy's overall profitability.

The main trade-off would be sacrificing part of the **positive convexity** exhibited by the strategy during the most severe market dislocations (2009, 2020 and 2025). A lower leverage cap would naturally reduce the contribution of these rare outliers, while still preserving the core mean-reversion mechanism of the framework.

For this reason, the leverage cap should primarily be viewed as a **risk-management parameter** rather than the source of the strategy's edge. Investors with lower risk tolerance can tailor the maximum leverage to their own preferences, recognizing that, while a lower cap would have reduced returns over the historical sample by limiting participation in a handful of exceptionally profitable stress episodes, it could also **improve the strategy's resilience to future market environments characterized by more severe or less mean-reverting tail events**.

Risk Premium or Market Inefficiency?

We do not view the strategy's returns as risk-free arbitrage. Our interpretation is that the observed **edge comes from investor behavior during stress periods**. When equity markets weaken, many participants become less willing to bear risk: they cut exposure, seek downside protection, or are forced to rebalance.

As discussed earlier, this can create a **feedback loop** in which selling pressure, hedging demand, and volatility-sensitive deleveraging reinforce one another: the combined effect may push equity prices below what fundamentals alone would justify.

By identifying these volatility bursts, the strategy is effectively **betting that part of the move is temporary and likely to be absorbed** over the following sessions.

Conclusion

We believe the main takeaway of this article is that **implied volatility can be used not only as a tradable asset class, but also as a behavioral stress signal** for timing equity exposure.

In our tests, a simple volatility-burst framework applied to SPY produced a 10.8% CAGR net of transaction costs (broadly in line with SPY buy-and-hold over the same sample), yet with **notable capital efficiency**, being invested on only 14% of trading days.

Importantly, the **parameters have not been optimized**, and we leave that fragile exercise to interested readers. Our goal was to show that even a simple use of implied-volatility information can already produce an interesting equity-timing signal.

There is **clear room for improvement**.

For example, the **sizing rule** is intentionally basic: exposure increases in discrete steps and is only loosely tied to the magnitude of the volatility burst. More adaptive methods could make the strategy less linear and more robust, with position sizing dependent on realized volatility, drawdown, liquidity, or even option-implied skew.

A second natural extension is to test **whether the logic holds beyond SPY**. The VIX is the most widely followed implied-volatility index, but only one member of a broader family. Natural pairs include QQQ-VXN, IWM-RVX, and DIA-VXD, and the framework could extend to sector ETFs, commodities, or large single names wherever a reliable index exists (such as GVZ for gold, OVX for crude oil, or VXAPL for Apple).

Regardless of the pair being studied, we believe the core idea remains portable: use implied volatility as a behavioral stress signal, then trade the underlying instrument **rather than the volatility product itself**.

If you found this research useful, feel free to share it with colleagues or friends who are interested in systematic trading, volatility, and short-term equity strategies.

Share

Disclaimer: While this study presents a comprehensive analysis of a systematic equity trading strategy based on implied-volatility signals, all results are derived from historical backtests. As such, they reflect past market behavior and should not be interpreted as guarantees of future performance. The framework discussed in this article relies on the empirical observation that episodes of elevated implied volatility

are, on average, followed by short-term mean reversion in equity prices. There is no assurance that this relationship will persist under future market conditions, particularly during structural market changes or prolonged periods in which extreme downside moves fail to revert as they have historically. Readers and investors should therefore view the findings as evidence from a historical sample rather than as a prediction of future returns.

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